

Hugging Face Transformers — Six NLP Tasks, Suite sentiment, NER, QA Translation search with pipeline

Overview

This project is a hands-on survey of six core natural language processing tasks solved using Hugging Face's pipeline API: sentiment classification, text summarization, named entity recognition, extractive question answering, machine translation, and semantic search. Every task uses a pretrained, open-weights model downloaded from Hugging Face Hub, requiring no fine-tuning, no training loop, and no API key.

What the Notebook Does

Sentiment classification is tested across a genuine positive movie review, a rewritten negative version of the same review, and a neutral sentence, covering the full sentiment spectrum rather than a single example. Summarization compares an explicit length constraint against the pipeline's default on both a long editorial and a shorter email. Named entity recognition is tested on deliberately ambiguous sentences to confirm the model distinguishes Apple the company from apple the fruit using context. Extractive question answering is tested against real passages, including one deliberately unanswerable question, confirming the model can recognise when no valid answer exists in the given context rather than fabricating one, since extractive QA selects a text span rather than generating free text.

Translation demonstrates the same one-line pipeline pattern applied to a full sequence-to-sequence task. The notebook's most substantial section builds a semantic search engine from first principles using sentence-transformers: ten documents are embedded into a shared vector space, a query is embedded on demand, and cosine similarity with torch's top-k function retrieves the most relevant match, wrapped into a reusable function and tested against two queries phrased nothing like the stored documents' exact wording, confirming genuine semantic rather than keyword matching.

Skills Demonstrated

- Rapid NLP prototyping using the Hugging Face pipeline abstraction
- Task-appropriate model selection across classification, generation, and extraction
- Building semantic search from raw sentence embeddings and cosine similarity
- Honest failure testing for extractive question answering

This multi-task transformer survey and the from-scratch semantic search implementation directly inform the retrieval components of my dissertation's hybrid GraphRAG and HippoRAG agentic intelligence platform.